EcoWatt AI – Intelligent Energy Consumption Prediction and Optimization Platform

Team Members

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Project Overview

With increasing electricity demand, rising energy costs, and growing environmental concerns, efficient energy management has become essential for sustainable development. Many households, educational institutions, and small businesses consume electricity without understanding their consumption patterns, resulting in unnecessary energy waste, higher electricity bills, and increased carbon emissions.

EcoWatt AI is an intelligent energy optimization platform that leverages Artificial Intelligence, Machine Learning, and Data Analytics to monitor, analyze, predict, and optimize electricity consumption. The platform processes historical energy usage data to identify consumption trends, forecast future electricity demand, detect abnormal energy usage, estimate carbon emissions, and calculate an overall Energy Efficiency Score.

Beyond prediction, EcoWatt AI provides AI-powered personalized energy-saving recommendations and classifies users based on consumption patterns using Machine Learning. By transforming raw energy data into actionable recommendations, the platform empowers users to reduce electricity costs, improve energy efficiency, and contribute to a more sustainable future.

Key Features

- AI-Based Electricity Consumption Prediction
- Energy Efficiency Score Analysis
- Personalized AI Energy Saving Recommendations
- Interactive Energy Analytics Dashboard
- Appliance-wise Energy Consumption Analysis

- Machine Learning-Based Energy Usage Classification
- Consumer Clustering Based on Energy Patterns
- Electricity Cost Prediction
- Carbon Emission Estimation
- Monthly & Yearly Energy Consumption Trends
- Abnormal Energy Usage Detection
- Personalized Energy Optimization Report

Technologies Used

- Python
- Pandas & NumPy
- Matplotlib, Seaborn & Plotly
- Scikit-learn
- Git & GitHub

Artificial Intelligence Components

Machine Learning

- Electricity Consumption Prediction using Linear Regression
- Energy Usage Classification using Random Forest / Decision Tree
- Consumer Segmentation using K-Means Clustering

Data Analytics

- Energy Consumption Analysis
- Electricity Cost Analysis
- Carbon Emission Analysis
- Energy Trend Forecasting

Recommendation Engine

- AI-generated personalized energy-saving recommendations based on user consumption patterns.

Sustainable Development Goals (SDGs)

This project contributes to multiple United Nations Sustainable Development Goals:

- **SDG 7 –** Affordable and Clean Energy
- **SDG 11 –** Sustainable Cities and Communities
- **SDG 12 –** Responsible Consumption and Production
- **SDG 13 –** Climate Action
- **SDG 17 –** Partnerships for the Goals

Expected Outcome

Develop an intelligent AI-driven energy management platform capable of predicting future electricity consumption, identifying inefficient energy usage patterns, estimating electricity costs and carbon emissions, and generating personalized energy-saving recommendations. The platform enables households, educational institutions, and organizations to make data-driven decisions that improve energy efficiency, reduce operational costs, and support sustainable energy consumption while contributing to global climate action.